



DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Title: Configuration of OSPF

COMPUTER NETWORKING LAB
CSE 312



GREEN UNIVERSITY OF BANGLADESH

1 Objective(s)

- To build and design a network using Cisco Packet Tracer
- To learn about step-by-step configuration of OSPF using Cisco Packet Tracer

2 Problem analysis

OSPF (Open Shortest Path First) is a dynamic routing protocol used in computer networks to efficiently route data packets within a single autonomous system (AS).

For example, imagine a large corporate network with multiple departments interconnected by routers. OSPF helps these routers communicate by maintaining a database of network topology, including link-state advertisements (LSAs) that describe the state and cost of links between routers.

If a link within the network fails, OSPF quickly recalculates routes based on the remaining available paths, ensuring minimal disruption to communication. This fast convergence time is a key feature of OSPF, crucial for maintaining network efficiency and reliability.

Moreover, OSPF's hierarchical design allows administrators to divide the network into areas, each with its own routing information. This helps manage scalability and reduces routing traffic, improving overall network performance.

OSPF plays a vital role in large-scale networks, such as corporate infrastructures or internet service provider networks, by providing efficient and reliable routing capabilities. There are some roles of router in OSPF:

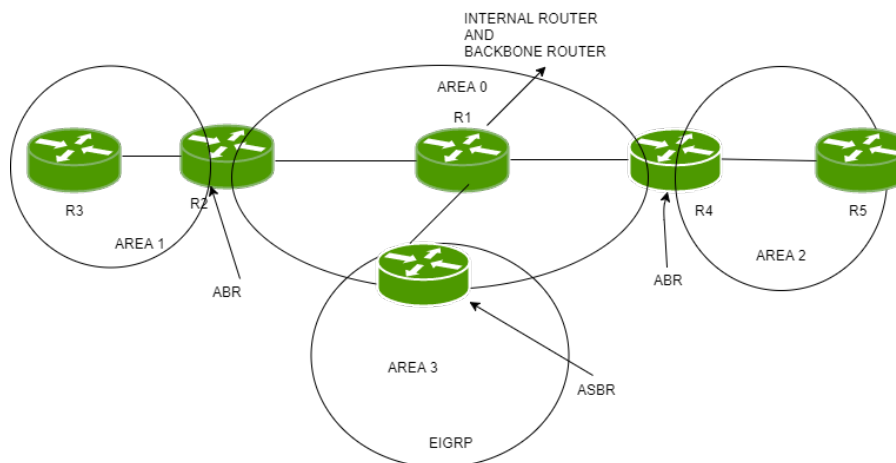


Figure 1: OSPF.

- **Backbone router**— The area 0 is known as backbone area and the routers in area 0 are known as backbone routers. If the routers exist partially in the area 0 then also it is a backbone router.
- **Internal router** – An internal router is a router which has all of its interfaces in a single area.
- **Area Boundary Router (ABR)** – The router which connects backbone area with another area is called Area Boundary Router. It belongs to more than one area. The ABRs, therefore, maintain multiple link-state databases that describe both the backbone topology and the topology of the other areas.
- **Area Summary Border Router (ASBR)** – When an OSPF router is connected to a different protocol like EIGRP or Border Gateway Protocol, or any other routing protocol then it is known as AS. The router which connects two different AS (in which one of the interfaces is operating OSPF) is known as Area Summary Border Router. These routers perform redistribution. ASBRs run both OSPF and another routing protocol, such as RIP or BGP. ASBRs advertise the exchanged external routing information throughout their AS.

So, in figure 2, we are going to take four Routers where one PC is connected to each. Router 1 and Router 2 belong to Area 1 and Router 3 and Router 4 belong to Area 2. OSPF Area Configuration we are going to do while configuring the route. Now, let's add IP Address on all the devices according to the picture given above.

3 Configuration

1. Build the network topology

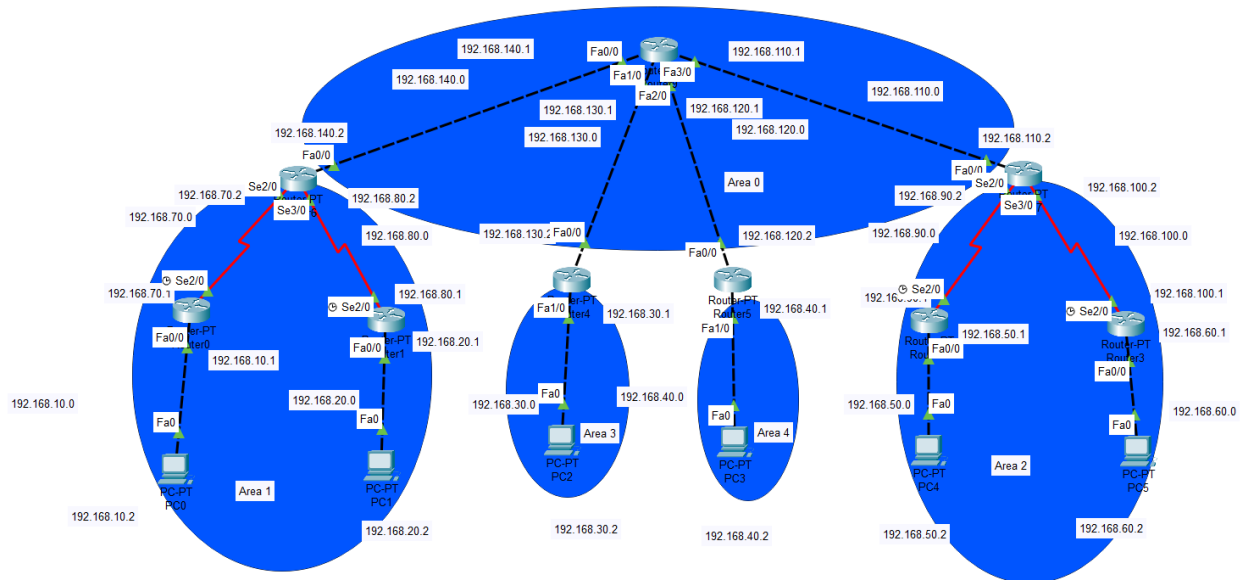


Figure 2: Build the network for OSPF Configuration.

2. Configure IP addresses on the PC and Router(Figure 3).

a) Click the device and go to the **Desktop tab > IP Configuration**.

For PC0: IP: 192.168.10.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.10.1

For PC1: IP: 192.168.20.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.20.1

For PC2: IP: 192.168.30.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.30.1

For PC3: IP: 192.168.40.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.40.1

For PC4: IP: 192.168.50.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.50.1

For PC5: IP: 192.168.60.2 Subnet Mask: 255.255.255.0 Gateway: 192.168.60.1

b) Click the router and go to the **Config > Interface**.

3. **OSPF Routing Commands.**

(a) **Router ospf 10**

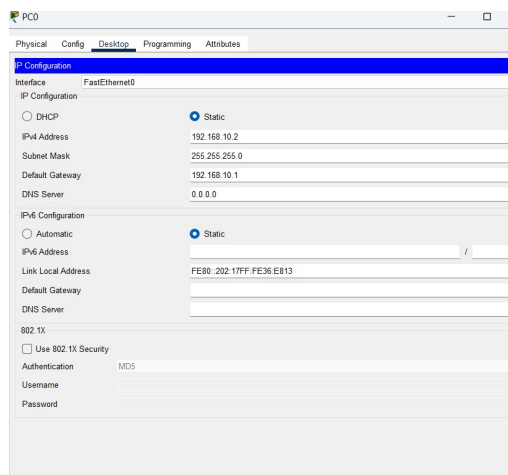
This command enables the OSPF protocol on your router, with 10 as the process ID. You can assign a process ID between 1 and 65,535.

(b) **Network 192.168.10.0 0.0.0.255 area 1.**

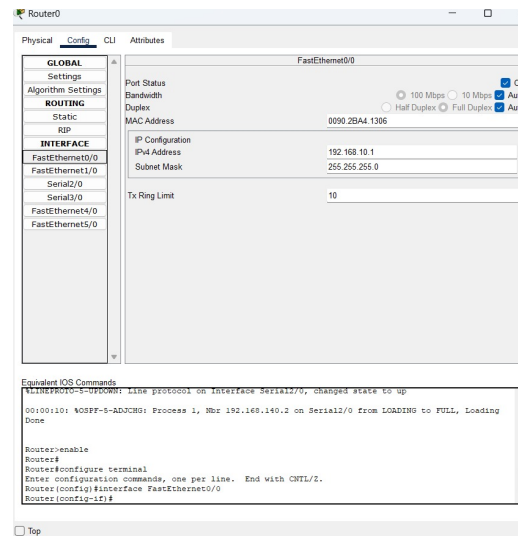
This command creates a route in OSPF, where '192.168.10.0' is the Network ID that the router is connected to in area 1. Here, '0.0.0.255' is the Wildcard Mask, which is the inverse of the Subnet Mask '255.255.255.0'. The wildcard mask is calculated by subtracting the subnet mask from '255.255.255.255', resulting in '0.0.0.255'.

(c) In the end, we specify 'area 1'. In OSPF, the area defines a separate network boundary. All routers within the same area can communicate directly, but different areas require additional configuration for communication.

4. OSPF Configuration Commands Step by Step.



(a) IP Configuration of PC



(b) IP configuration of router

Figure 3: IP Configuration of End Devices and Router

(a) Configure OSPF in Router 0

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.70.0 0.0.0.255 area 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 1
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

(b) Configure OSPF in Router 1

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.80.0 0.0.0.255 area 1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 1
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

(c) Configure OSPF in Router 4

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.130.0 0.0.0.255 area 3
Router(config-router)#network 192.168.30.0 0.0.0.255 area 3
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

(d) Configure OSPF in Router 5

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.120.0 0.0.0.255 area 4
Router(config-router)#network 192.168.40.0 0.0.0.255 area 4
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

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- (e) Configure OSPF in Router 2

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.90.0 0.0.0.255 area 2
Router(config-router)#network 192.168.50.0 0.0.0.255 area 2
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

- (f) Configure OSPF in Router 3

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.100.0 0.0.0.255 area 2
Router(config-router)#network 192.168.60.0 0.0.0.255 area 2
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

- (g) Configure OSPF in Router 6

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.140.0 0.0.0.255 area 0
Router(config-router)#network 192.168.70.0 0.0.0.255 area 1
Router(config-router)#network 192.168.80.0 0.0.0.255 area 1
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

- (h) Configure OSPF in Router 7

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.110.0 0.0.0.255 area 0
Router(config-router)#network 192.168.90.0 0.0.0.255 area 2
Router(config-router)#network 192.168.100.0 0.0.0.255 area 2
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

- (i) Configure OSPF in Router 8

```
Router # en
Router#conf t
Router(config)#router ospf 10
Router(config-router)#network 192.168.140.0 0.0.0.255 area 1
Router(config-router)#network 192.168.110.0 0.0.0.255 area 2
Router(config-router)#network 192.168.130.0 0.0.0.255 area 3
Router(config-router)#network 192.168.120.0 0.0.0.255 area 4
Router(config-router)# Z (Use Ctrl + Z to save settings)
```

4 Input/Output

After the successful compilation of ospf configuration, we simply test our configuration by ping command.

5 Discussion & Conclusion

Based on the focused objective(s) to learn the step-by-step configuration of an OSPF. This task will help students learn the principles of networking with hands-on experience as well as develop Cisco technology-specific skills. The additional lab exercise will help them to practice OSPF configuration and also help them to be confident towards the fulfillment of the objectives(s).

6 Lab Task (Please implement yourself and show the output to the instructor)

Implement OSPF for the following topology

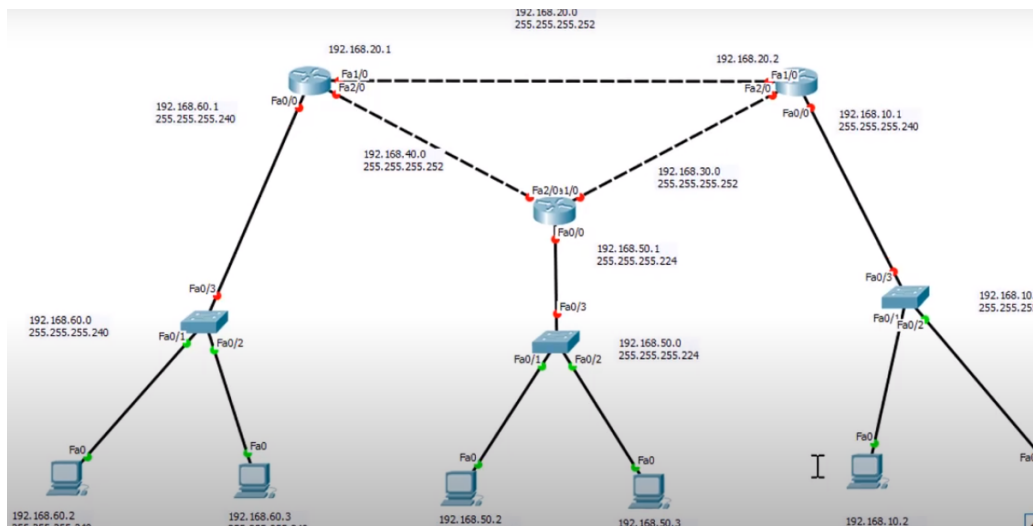


Figure 4: OSPF

7 Lab Exercise (Submit as a report)

You will work on a simple network topology consisting of:

- 4 routers (R1, R2, R3, R4)
- 3 switches (SW1, SW2, SW3)
- 6 PCs (PC1, PC2, PC3, PC4, PC5, PC6)

Requirements:

1. Configure IP addresses on all interfaces of routers and PCs according to the provided IP addressing scheme
2. Enable OSPF routing protocol on routers and ensure full connectivity between all PCs.
3. Verify OSPF neighbor adjacencies and routing tables.
4. Troubleshoot any OSPF-related issues that may arise during the configuration.

8 Policy

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